Healthcare Patient Management System — API Specification

Complete RESTful Endpoints, Request Schemas, Authorization Roles, and Responses

Endpoint & Method	Roles	Description	Sample Payload / Params
POST /auth/login	Public	User authentication returning JWT bearer token	username, password
GET /auth/me	All Authenticated	Fetch current authenticated user profile	Bearer Token
GET /patients	Admin, Staff, Analyst	Paginated search & list patients with gender/race filters	limit=20, offset=0, search=...
POST /patients	Admin, Staff	Register a new patient master record	{"patient_nbr": 8881234, "race": "Asian", "gender": "Male"}
PATCH /patients/{id}	Admin, Staff	Update patient demographic information	{"gender": "Female"}
DELETE /patients/{id}	Admin	Cascade delete patient and associated encounters	Path parameter: id
GET /patients/{id}/encounters	Admin, Staff, Analyst	Retrieve full clinical admission timeline for patient	Path parameter: id
GET /encounters	Admin, Staff, Analyst	List inpatient admissions with filtering by readmission and age	readmitted=<30, age=[60-70)
POST /encounters	Admin, Staff	Create a new inpatient encounter record	{"encounter_id": 999, "patient_nbr": 8222, ...}
GET /analytics/overview	Admin, Staff, Analyst	Hospital KPIs (Total patients, encounters, readmission %, avg stay)	None
GET /analytics/admissions	Admin, Staff, Analyst	Encounter volumes and average stay grouped by admission type	None
GET /analytics/readmissions	Admin, Staff, Analyst	Readmission breakdown (<30, >30, NO) and risk distribution	None
GET /analytics/length-of-stay	Admin, Staff, Analyst	Encounter frequency distributed across LOS day brackets	None
GET /reports/export/encounters/csv	Admin, Staff, Analyst	Direct RFC 4180 CSV export of encounters dataset	limit=500
GET /reports/export/encounters/excel	Admin, Staff, Analyst	Microsoft Excel (.xlsx) formatted encounter export	limit=500
GET /reports/audit-logs	Admin Only	Retrieve compliance ledger of mutating actions	limit=50